

Data-Driven Multiobjective Multimodal Transportation Route Optimization in Hybrid Uncertain Environments

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Abstract—Multimodal freight transportation is essential for enhancing logistics efficiency and reducing costs, with route optimization as its core component. However, uncertainties are prevalent in multimodal systems, posing challenges for the construction and validation of simulation models. Furthermore, the involvement of multiple stakeholders introduces conflicting optimization objectives. Effectively balancing these objectives to maximize overall benefits has become a critical issue that requires resolution. For the multimodal transport route optimization problem in hybrid uncertainty environments (e.g., there are uncertainties in transportation demand, transportation time, and transfer time), this study constructs a multiobjective optimization model based on fuzzy numbers. The complexity of the model is reduced by introducing the chance-constrained programming theory. To solve the model, a data-driven multi-objective evolutionary algorithm is designed, integrating Monte Carlo simulation with surrogate models to effectively reduce the computational cost of uncertainty estimation. Furthermore, a constraint prioritization strategy is developed to handle multiple conflict objectives, and complex constraints efficiently. Simulation results demonstrate that the proposed algorithm exhibits excellent performance across networks of varying scales, providing robust decision support for multimodal transportation decision-making.

Index Terms—Data-driven optimization, multiobjective optimization, multimodal transportation, chance-constrained programming, evolutionary algorithms, hybrid uncertain environments.

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I. INTRODUCTION

MULTIMODAL transportation is a logistics mode that integrates various means of transportation such as land, rail, sea, and air [1]. This mode implements standardized management of the entire transport process through a single transport contract and uniform transport vouchers, realizing the integration of the transport process. Its core advantage lies in the optimization of the combination of different modes of transport to reduce transport costs and improve transport efficiency. Furthermore, multimodal transportation accounts for both the flexibility of transportation and environmental sustainability, which has led to its widespread application in international trade and domestic logistics sectors [2].

The multimodal transportation multiobjective routing optimization problem is widely regarded as very challenging in both academic and practical fields [3]. The core difficulty lies in the need to make delicate trade-offs between a series of conflicting optimization objectives, including minimizing transportation costs, reducing transportation times, and mitigating environmental impacts (such as minimizing carbon emissions). The inherent conflicts among these objectives make achieving a globally optimal solution under specific conditions exceptionally complex. Additionally, external uncertainties such as fluctuations in transportation demand and the unpredictability of transportation times further exacerbate the difficulty of route optimization. These uncertainties require decision-makers to comprehensively assess potential risks and variables and possess the capability to flexibly adjust strategies in a dynamically changing environment. This not only necessitates a deep understanding of existing transportation modes but also demands that decision-makers employ scientific methods and tools to achieve an optimal balance between transportation efficiency and cost-effectiveness, thereby seeking comprehensive solutions within the complex system of multimodal transportation.

The core challenge of multimodal transportation route planning lies in the need to develop and evaluate a series of alternative solutions to provide decision support and risk management reference for logistics enterprises [4], [5]. This process involves extensive participation from various stakeholders and requires a deep understanding of the complexity and uncertainty of the external environment. A single route plan often fails to meet multidimensional transport demands [6]. Initial plans are proposed by the primary carrier, but final plans

TABLE I
A SUMMARY OF THE LITERATURE ON DATA-DRIVEN OPTIMIZATION

Study	Research question	Methodology
Wang et al. [13] (2019)	How to leverage historical data to guide evolutionary optimization in an offline setting without new data	Construct a selective radial basis function surrogate ensemble and use an adaptive subset-selection strategy for fitness estimation
Peng et al. [14] (2023)	How to jointly optimize cost and time under uncertain travel time while improving computational efficiency	Build a stochastic multiobjective model; employ an ant colony algorithm combined with a data pool and a support vector regression surrogate to accelerate simulation
Lou et al. [15] (2021)	How to automatically optimize shear-wall layouts in high-rise buildings under structural performance constraints, reducing weight and improving design efficiency	Discretize shear walls into binary encoding; use an improved tabu search combined with conceptual design rules and a support vector machine surrogate model
Wang and Jin [16] (2020)	How to handle expensive objective and constraint functions in multiobjective combinatorial optimization that relies on historical data	Use a random-forest surrogate to approximate objectives and constraints; apply logistic regression to correct constraint boundaries with stochastic ranking
Jiang et al. [17] (2022)	How to reduce code size and lower the cost of expensive fitness evaluations in compiler optimization sequence selection	Embed local search into a genetic algorithm and use a random-forest surrogate so that predictions can substitute for actual compilation evaluations

must be determined through collaboration and negotiation among multiple stakeholders. The multimodal transportation process follows established rules, with stakeholders conducting in-depth discussions on specific transport routes before finalizing the plan. This study offers multiple plans to provide decision-makers with diverse options to deal with interference from external factors. For instance, certain departments may object to the transit of freight through specific areas, rendering specific routes or nodes in the original plan unfeasible. Consequently, the transport plan must be adjusted, making it crucial to provide multiple alternative solutions. By comprehensively considering multiple optimization objectives such as cost, time, and environmental impact, several feasible solutions are proposed. Decision-makers can select the most appropriate solution based on the specific transport environment and personal preferences, thereby enhancing decision-making efficiency and adaptability. Additionally, the process from plan formulation to implementation requires time, and the complexity and variability of the transport environment may affect plan execution. Therefore, providing multiple solutions is essential.

Evolutionary algorithms are widely employed for multiobjective optimization problems (MOPs) due to their remarkable flexibility and robustness, which enable them to effectively estimate the Pareto optimal set in a single run [7]. These algorithms, known as multiobjective evolutionary algorithms (MOEAs), utilize population-based search mechanisms to accurately estimate the non-dominated solution set and reveal the trade-offs between conflicting objectives [8], [9]. MOEAs (e.g., NSGA-II [10]) are widely used when solving multiobjective optimization problems. MOEAs benefit from their flexible search mechanism, which usually assumes that the inputs, constraints, and objective function of the problem are deterministic. However, uncertainty is very common in real applications, which leads to the performance of metaheuristic algorithms dramatically degrades. An effective method is to extend the MOEAs through simulation technology to cope with uncertain scenarios. Monte Carlo simulation is grounded in probabilistic statistical theory and extensively employs random numbers to model random variables. It derives numerical solutions to problems by using statistical data characteristics

obtained through sampling, such as mean, variance, or probability [11]. When appropriately integrated with evolutionary algorithms, Monte Carlo simulation has demonstrated broad effectiveness in engineering practice [12].

However, in data-driven evaluations of objectives and constraints, the process often relies on computationally expensive numerical simulations. Fortunately, the introduction of data-driven strategies in simulation optimization algorithms can significantly reduce the computational complexity [13]. For example, Peng et al. [14] proposed a multiobjective optimization model for solving multimodal transportation routing problems with stochastic transportation time. They utilized a data-driven multiobjective ant colony algorithm to accelerate the process of finding non-dominated solutions. Lou et al. [15] introduced a high-rise building shear wall layout optimization strategy that combines conceptual design with data-driven tabu search. This method uses support vector machines to develop surrogate models that drive the optimization process, thereby reducing the cost of evaluating actual models. Wang et al. [13] proposed a novel offline data-driven evolutionary algorithm that employs an ensemble learning-based surrogate management strategy. By constructing a large number of surrogate models and adaptively selecting a subset, this approach achieves optimal local approximation accuracy and reduces computational complexity. Wang and Jin [16] developed a random forest-assisted evolutionary algorithm for solving data-driven constrained multiobjective combinatorial optimization problems, particularly suitable for expensive, computation-intensive issues such as trauma system design. Overall, incorporating historical data into the iterative process of evolutionary algorithms helps alleviate the high time and cost burdens associated with expensive simulations [17]. Finally, Table I summarizes the literature on data-driven optimization.

Motivated by the above considerations, this study focuses on the practical operational context of multimodal transportation and addresses the combined uncertainties of factors such as transportation demand, transportation time, and transfer time within a multimodal transport network. A multiobjective route optimization model with fuzzy chance constraints is developed. To solve the model, Monte Carlo simulation is

utilized to generate sample datasets, allowing for the addressing of uncertain variables and the approximation of objective functions. Nevertheless, due to the high computational cost of generating large sample sets, a data-driven evolutionary optimization method is introduced to improve solution efficiency. The main contributions of this study are summarized as follows:

- 1) We consider a hybrid uncertain environment in which freight demand, transportation time, and transshipment time are all uncertain, and construct a tri-objective routing model integrating transportation cost, transportation time, and carbon emissions.
- 2) A data-driven multiobjective evolutionary algorithm with constraint prioritization is proposed to solve the model. Specifically, given the difficulty in accurately assessing uncertain variables within the model, Monte Carlo simulation is employed for sampling to manage uncertainty, though this approach can be computationally intensive. To this end, we train surrogate models based on historical data to predict uncertain variables, significantly reducing the computational burden of uncertainty estimation. Additionally, to handle multiple conflicting objectives and complex constraints within the model, a constraint prioritization technique is developed that gradually incorporates constraints into the evolutionary process. This technique helps to reduce the solution difficulty and effectively utilize potential infeasible solutions.
- 3) In the simulation experiments, the effectiveness of the proposed algorithm is validated on small, medium, and large-scale multimodal transportation test networks. Compared to five state-of-the-art evolutionary algorithms, the proposed approach shows significant advantages. Ultimately, the model and algorithm developed in this study efficiently generate multiple multimodal transportation solutions, offering robust decision support for multimodal transportation operators.

The remaining sections of the article are organized as follows: Section II outlines the research problem, basic model, and chance-constrained programming model. Section III details the implementation of the data-driven multiobjective evolutionary algorithm with constraint prioritization. Section IV presents experimental results to validate the proposed algorithm's effectiveness. Section V concludes with a summary and key findings.

II. PROBLEM AND MODEL

A. Problem Description

In this study, we consider a task involving freight transportation between two locations, where the demand, transportation time, and transshipment time are all uncertain. The transportation route must pass through multiple nodes, with various transportation modes available between adjacent nodes, including road, rail, and waterway. Given the differences in cost, time efficiency, and environmental impact among these transportation modes, this paper aims to explore a set of optimal transportation routes and modes that simultaneously optimize

transportation cost, transportation time, and carbon emissions under the constraints of mixed time windows, while comprehensively considering these multiple sources of uncertainty.

In practical multimodal transportation operations, to ensure that the research case closely reflects real-world scenarios, the following assumptions are made:

- 1) The state attributes of the freight remain unchanged, meaning any potential loss or damage during transportation is not considered;
- 2) If the modes of transportation for freight entering and exiting a transshipment node are the same, no transshipment operations are performed;
- 3) Freight is only transshipped at nodes, with a maximum of one transshipment allowed per node;
- 4) For each pair of nodes, only one transportation mode and one route are allowed to be selected.

B. Analysis of Uncertainties

In this study, we construct an uncertainty model involving multiple uncertain factors, including uncertain freight demand, uncertain transportation time, and uncertain transshipment time. Motivated by the problem's characteristics, we model demand and time using two types of fuzzy numbers within a single optimization model. Specifically, freight demand is typically influenced by market fluctuations, seasonal variations, and unexpected events. These factors cause the demand to cluster around a specific most likely value, with the likelihood of demand rapidly decreasing as it deviates from this value. Therefore, we choose triangular fuzzy numbers to describe this uncertain freight demand. The triangular fuzzy number $\tilde{q} = (q^a, q^b, q^c)$ is centered around the most likely value q^b , exhibiting symmetry, which allows it to accurately capture the concentration trend and uncertainty of the demand.

Furthermore, the uncertainty in transportation time and transshipment time is influenced by various factors such as traffic conditions, weather changes, and modes of transportation. These factors result in transportation time fluctuating within a larger interval. Trapezoidal fuzzy numbers are chosen to describe this uncertainty as they can effectively depict the stability of time within a high-probability interval and gradually decrease outside this interval. The trapezoidal fuzzy number $\tilde{t} = (t^a, t^b, t^c, t^d)$ consists of t^a representing the fuzzy left lower bound, t^b representing the fuzzy left upper bound, t^c representing the fuzzy right upper bound, and t^d representing the fuzzy right lower bound. This configuration can more accurately capture and describe the uncertainty of transportation time, particularly in actual transportation time predictions where time data typically exhibit a high-probability interval, such as peak periods or specific traffic segments. Trapezoidal fuzzy numbers can better reflect these high-probability intervals.

C. Notation Definition

In the study of multimodal transportation network models, the route optimization problem is modeled as a directed graph $G(V, E, M)$. In this model, V represents the set of all nodes, including the departure O and the destination D ; A denotes

TABLE II
GLOSSARY OF NOTATIONS IN MULTIMODAL TRANSPORTATION

Notations	Meaning
Set	
G	Multimodal transportation network
V	Set of nodes
A	Set of arcs
M	Set of transportation modes
Parameters	
C_{ij}^k	Unit transportation cost of the k -th transportation mode between (i, j)
d_{ij}^k	Transportation distance between (i, j) for the k -th mode
C_i^{kl}	Unit transshipment cost of the transportation mode switching from k to l at node i
p_1	Unit warehousing cost for early arrival
p_2	Unit penalty cost for delayed arrival
t_i	Time of arrival of freight at node i
E_{ij}^k	Unit carbon emissions of the k -th mode of transportation between (i, j)
E_i^{kl}	Unit carbon emissions of the transportation mode switching from k to l at node i
t_{ij}^k	Transportation time between (i, j) for the k -th mode
t_i^{kl}	Unit transshipment time of the transportation mode switching from k to l at node i
v_k	Average speed of transportation mode k
t_0	Departure time of transportation
t_{a_i}	Arrival time at node i
t_{d_i}	Departure time at node i
$\tilde{t}_{ij}^k$	Transportation time between (i, j) using mode k
$\tilde{t}_i^{kl}$	Transit time at node i mode from k to l
T_{min}	Minimum time for freight to arrive at destination
T_{max}	Maximum time for freight to arrive at destination
Swl_i	Soft time window lower bound at node i
Swu_i	Soft time window upper bound at node i
Hwl_i	Hard time window lower bound at node i
Hwu_i	Hard time window upper bound at node i
$\tilde{q}$	Fuzzy freight volume demand, $\tilde{q} = (q^a, q^b, q^c)$, q^a and q^c denote the lower and upper bounds of fuzzy freight volume respectively, q^b is the fuzzy freight volume with a membership degree of 1.
q	Freight volume transported
q_{ij}^k	Maximum transportation capacity between (i, j) for the k -th mode of transportation
q_i^{kl}	Maximum transshipment capacity of mode switching from k to l at node i
C_{max}	Total cost budget
E_{max}	Total carbon emissions threshold
Variables	
x_{ij}^k	If the k -th mode of transportation operates between (i, j) , then $x_{ij}^k = 1$; otherwise it is 0.
y_i^{kl}	If the modal shift at node i is from k to l , then $y_i^{kl} = 1$; otherwise, it is 0.

the set of all arcs, which connect different nodes and form the skeleton of the network; M represents the set of transportation modes, encompassing various modes of transport such as rail, road, and waterways. Table II provides a detailed explanation of these notations and their roles within the model, thereby laying the theoretical foundation for understanding and analyzing multimodal transportation networks.

D. Basic Multiobjective Optimization Model

This study develops a multiobjective optimization model aimed at minimizing total cost, total transportation time, and total carbon emissions during the freight transportation process. The model is designed to fully account for the multiple mixed uncertainty factors in the transportation system, including freight volume, transportation time, and transshipment time, thereby reflecting the complexity and dynamic nature of the transportation process. Additionally, the model considers

the mixed time windows for cargo delivery to various nodes, ensuring that transportation tasks are efficiently completed within the specified time frame. This enhances the model's practicality and its adaptability to different transportation scenarios. Overall, a basic multiobjective route optimization model for multimodal transportation is as follows:

$$\min F = [f_1, f_2, f_3] \quad (1)$$

$$\begin{aligned} f_1 &= f_C(\mathbf{x}, \mathbf{y}, \tilde{q}, \tilde{t}_{ij}^k, \tilde{t}_i^{kl}) \\ &= \tilde{q} \sum_{i \in V} \sum_{j \in V} \sum_{k \in M} C_{ij}^k d_{ij}^k x_{ij}^k \\ &\quad + \tilde{q} \sum_{i \in V} \sum_{k \in M} \sum_{l \in M} C_i^{kl} y_i^{kl} \\ &\quad + \tilde{q} p_1 \sum_{i=1}^V \max(Sw_{l_i} - t_{a_i}, 0) \\ &\quad + \tilde{q} p_2 \sum_{i=1}^V \max(t_{a_i} - Sw_{u_i}, 0) \end{aligned} \quad (2)$$

$$\begin{aligned} f_2 &= f_T(\mathbf{x}, \mathbf{y}, \tilde{q}, \tilde{t}_{ij}^k, \tilde{t}_i^{kl}) \\ &= \sum_{i \in V} \sum_{j \in V} \sum_{k \in M} \tilde{t}_{ij}^k x_{ij}^k \\ &\quad + \sum_{i \in V} \sum_{k \in M} \sum_{l \in M} \tilde{t}_i^{kl} y_i^{kl} \end{aligned} \quad (3)$$

$$\begin{aligned} f_3 &= f_E(\mathbf{x}, \mathbf{y}, \tilde{q}, \tilde{t}_{ij}^k, \tilde{t}_i^{kl}) \\ &= \tilde{q} \sum_{i \in V} \sum_{j \in V} \sum_{k \in M} E_{ij}^k d_{ij}^k x_{ij}^k \\ &\quad + \tilde{q} \sum_{i \in V} \sum_{k \in M} \sum_{l \in M} E_i^{kl} y_i^{kl} \end{aligned} \quad (4)$$

$$s.t. \quad f_1 \leq C_{max} \quad (5)$$

$$T_{min} \leq f_2 \leq T_{max} \quad (6)$$

$$f_3 \leq E_{max} \quad (7)$$

$$\tilde{q} x_{ij}^k \leq q_{ij}^k \quad (8)$$

$$\tilde{q} y_i^{kl} \leq q_i^{kl} \quad (9)$$

$$\sum_{i, j \in V, k \in M} x_{ij}^k - \sum_{j, h \in V, k \in M} x_{jh}^k = \begin{cases} 1, j = D \\ 0, j \in V \setminus \{O, D\} \\ -1, j = O \end{cases} \quad (10)$$

$$x_{ij}^k + x_{ij}^l \geq 2 y_i^{kl} \quad (11)$$

$$\sum_{k \in M} x_{ij}^k \leq 1 \quad (12)$$

$$\sum_{k \in M} \sum_{l \in M} y_i^{kl} \leq 1 \quad (13)$$

$$t_{a_i} = \begin{cases} 0, i = O \\ t_{d_i} + \sum_{i \in V} \sum_{k \in M} \tilde{t}_{ji}^k x_{ji}^k + \sum_{i \in V} \sum_{k, l \in M} \tilde{t}_i^{kl} y_j^{lk}, \\ \forall i \in V/O \end{cases} \quad (14)$$

$$t_{d_i} = t_{a_i} + \sum_{i \in V} \sum_{k \in M} \tilde{t}_{ij}^k x_{ij}^k + \sum_{i \in V} \sum_{k, l \in M} \tilde{t}_i^{kl} y_i^{kl}, \quad \forall i \in V/O \quad (15)$$

$$Hw_{l_i} \leq t_{a_i} \leq Hw_{u_i} \quad (16)$$

$$x_{ij}^k, y_i^{kl} \in \{0, 1\} \quad (17)$$

Eq. (1) describes the minimization of the three objective functions. Eq. (2) defines the total transportation cost, which covers transportation costs, transshipment costs, and penalty costs for violating soft time windows. Eq. (3) is the total transportation time, which consists of inter-nodal transportation time and transshipment time. Eq. (4) denotes the total carbon emission, which consists of the inter-nodal transportation carbon emission and the transshipment carbon emission. Eq. (5) limits the maximum budget for freight costs. Eq. (6) specifies the time window for the arrival of freight at the destination. Eq. (7) sets the maximum carbon emission threshold. Eqs. (8) and (9) are the constraints on the maximum transport capacity and the maximum transshipment capacity, respectively. Eq. (10) ensures the balance of freight volume at each node. Eq. (11) guarantees the continuity of multimodal transportation. Eq. (12) ensures that each route segment can only use a specific transportation mode and cannot be split. Eq. (13) stipulates that freight can be transshipped at most once at each node. Eqs. (14) and (15) respectively calculate the arrival and departure times of freight at node i . Eq. (16) represents the hard time window constraint for the arrival of the goods at node i . Eq. (17) defines the range of 0-1 decision variables.

E. Chance-Constrained Multiobjective Programming Model

Since the uncertain multimodal transportation route optimization model cannot be solved directly, it needs to be transformed into a deterministic form. This paper employs chance-constrained programming theory to transform the uncertainty model into a fuzzy chance-constrained model [18]. Specifically, the model discussed in Section II-D involves multiple mixed uncertain factors, which propagate uncertainty to both the objective function and the constraints.

According to the principles of fuzzy chance constraints, at a specified confidence level, the uncertain objective function and constraints can be transformed into their optimal possible values. This transformation not only clarifies the model theoretically but also enhances its practical applicability. The transformed model is as follows:

$$\min \bar{F} = [\bar{f}_1, \bar{f}_2, \bar{f}_3] \quad (18)$$

$$\text{s.t. } \Pr\{f_1(\mathbf{x}, \mathbf{y}) \leq \bar{f}_1\} \geq \alpha_1 \quad (19)$$

$$\Pr\{f_2(\mathbf{x}, \mathbf{y}) \leq \bar{f}_2\} \geq \alpha_2 \quad (20)$$

$$\Pr\{f_3(\mathbf{x}, \mathbf{y}) \leq \bar{f}_3\} \geq \alpha_3 \quad (21)$$

$$\Pr\{\tilde{q}x_{ij}^k \leq q_{ij}^k\} \geq \beta_1 \quad (22)$$

$$\Pr\{\tilde{q}y_i^{kl} \leq q_i^{kl}\} \geq \beta_2 \quad (23)$$

$$\Pr\{f_1(\mathbf{x}, \mathbf{y}) \leq C_{max}\} \geq \beta_3 \quad (24)$$

$$\Pr\{T_{min} \leq f_2(\mathbf{x}, \mathbf{y}) \leq T_{max}\} \geq \beta_4 \quad (25)$$

$$\Pr\{f_3(\mathbf{x}, \mathbf{y}) \leq E_{max}\} \geq \beta_5 \quad (26)$$

$$\alpha_1, \alpha_2, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5 \in (0, 1) \quad (27)$$

where $\Pr\{\cdot\}$ denotes the probability of the event in $\{\cdot\}$, and α and β are predetermined confidence levels.

In addition to the above equations, Eqs. (10)–(17) are also included and remain unchanged. Furthermore, all constraints

must be simultaneously satisfied within the same scenario:

$$\begin{aligned} &\Pr\{\tilde{q}x_{ij}^k \leq q_{ij}^k \\ &\&\tilde{q}y_i^{kl} \leq q_i^{kl} \\ &\&f_1(\mathbf{x}, \mathbf{y}) \leq C_{max} \\ &\&T_{min} \leq f_2(\mathbf{x}, \mathbf{y}) \leq T_{max} \\ &\&f_3(\mathbf{x}, \mathbf{y}) \leq E_{max}\} \geq \beta_6 \\ &\beta_6 \in (0, 1) \end{aligned} \quad (28)$$

III. DATA-DRIVEN MULTI-OBJECTIVE EVOLUTIONARY ALGORITHM WITH CONSTRAINT PRIORITIZATION

To address the constructed fuzzy chance-constrained multi-objective optimization model, this paper innovatively proposes a data-driven multiobjective evolutionary algorithm. The fuzzy chance-constrained multiobjective programming model outlined in Section II-E represents a classic example of expensive constrained multi-objective optimization problems [19], [20]. On the one hand, due to the high uncertainty in transportation demand, segment transportation time, and node transfer time, a large number of samples are required to accurately estimate the objective function, making the evaluation process costly. On the other hand, the model involves multiple conflicting objectives and complex constraints, which not only increase the difficulty of solving the problem but also make it challenging to obtain feasible solutions. To address these issues, we adopted a data-driven approach to reduce the computational cost associated with objective function evaluations. Additionally, to tackle the complexity of constraints and the difficulty in finding feasible solutions, we designed a constraint-handling technique (CHT) based on constraint prioritization. In the CHT, constraints are progressively introduced and handled at different evolutionary stages. Specifically, in the early stages of the algorithm, only a subset of constraints is considered, allowing the population to rapidly explore promising regions in the decision space. As the evolution progresses, additional constraints are gradually introduced to refine the solution set based on high-quality solutions, thereby balancing solution quality with computational cost.

A. Algorithm Framework

Algorithm 1 presents the pseudocode of the proposed data-driven and constraint-priority-based multiobjective evolutionary algorithm (i.e., DD-CPEA). Initially, the algorithm randomly initializes N solutions to form the main population and archive population, followed by evaluating these solutions (lines 1-3). Then, enter the iteration, where constraint priorities are dynamically determined based on the feasibility ratio of non-dominated solutions in the main population (line 6), with further details outlined in Algorithm 2. Subsequently, the main population is randomly re-initialized. Lines 8-22 outline the process for determining the criteria under which the algorithm transitions between evolutionary stages. If the criteria are satisfied, the algorithm progresses to the next stage, introducing new constraints based on their priority and replacing the current population with an expanded archive population. Otherwise, the population remains in

Algorithm 1 Framework of DD-CPEA

Input: N (population size), $MaxFEs$ (maximum number of function evaluations)

Output: P (main population)

```

1:  $P \leftarrow$  Randomly initialize  $N$  solutions
2:  $A \leftarrow P$  (archive population)
3: Evaluate  $P$  by MOP
4: while termination condition is not met do
5:   /* Algorithm 2 */
6:    $Prio \leftarrow$  Constraint_Prioritization( $P, \Delta$ )
   /*Algorithm 2*/
7:    $P \leftarrow$  Randomly reinitialize  $N$  solutions
8:    $current\_cons \leftarrow \emptyset$ 
9:   if  $|current\_cons| = 0$  then
10:     $fr \leftarrow$  Feasibility ratio of  $P$  on highest-priority
    constraint
11:    if  $fr < 1$  then
12:      /* Enter the next stage and add a constraint*/
13:       $new\_con \leftarrow$  Dequeue( $cons\_queue, Prio$ )
14:       $current\_cons \leftarrow current\_cons \cup new\_con$ 
15:    else
16:      if stage transition criteria is met then
17:         $new\_con \leftarrow$  Dequeue( $cons\_queue, Prio$ )
18:         $current\_cons \leftarrow current\_cons \cup new\_con$ 
19:         $P \leftarrow A$ 
20:      end if
21:    end if
22:  end if
23:   $O \leftarrow$  Search_Operator( $P, N$ ) /*Algorithm 3*/
24:   $P \leftarrow P \cup O$ 
25: Evaluate  $P$  by  $current\_cons$  and
26: data-driven method in Section III-B
27: if  $|current\_cons| > 0$  then
28:   /* Algorithm 3 */
29:    $A \leftarrow$  Archive_Update( $[A, O], N, current\_cons$ )
30: end if
31:  $P \leftarrow$  Environmental_Selection ( $P, N, current\_cons$ )
32: end while
33: Return  $P$ 

```

the current stage. In the next steps, offspring are generated using reproduction operators, and the main population is merged with the offspring to form the new main population (lines 23-24). The new population is then evaluated using the constraints of the current stage and the data-driven method described in Section III-B (lines 23-24). Algorithm 3 updates the archive population based on the current stage's constraints and the newly generated offspring (lines 27-30). Finally, environmental selection is applied to the main population according to the constraints of the current stage (line 31). The algorithm returns the set of multimodal transportation routes corresponding to the non-dominated solutions in the main population as the final result.

B. Data-Driven Method

1) *Random Forest*: Most surrogate models are designed to simulate and approximate continuous functions. However,

decision variables in combinatorial optimization problems are inherently discrete in nature. To this end, this study utilizes random forest models as surrogates to approximate uncertain transportation demand, transportation time, and transfer time. Recent studies have demonstrated that the tree-based structure of random forests provides significant advantages in approximating functions with discrete decision variables [21], [22].

The random forest model is an ensemble of classification and regression trees (CARTs) [23]. The binary tree structure of CARTs makes random forests well-suited for modeling discrete problems. A CART divides the decision space recursively into rectangular regions defined by leaf nodes, where the output value of each leaf node corresponds to the mean of the sample outputs within each divided region. The growth mechanism of CART is governed by specific splitting and stopping criteria, with the core objective being to minimize the mean squared error (MSE) of the tree. Specifically, CART halts its growth when the reduction in MSE from further splits falls below a predefined threshold of $1 \times 10^{-4} \times \sigma^2$ (where σ^2 represents the overall variance of the dataset). As illustrated in Fig. 3, each CART in a random forest model is trained on different bootstrap samples, resulting in the diversity of tree structures depicted in Fig. 3. When dealing with problems with n training samples involving uncertain demand or transportation time, each CART randomly selects $d = 2\lfloor \sqrt{D} \rfloor$ decision variables from a total of D variables to compose its n bootstrap samples. The selection mechanism allows a variable to be selected multiple times within a tree. The final prediction y is obtained by averaging the predictions from k decision trees. In this study, following the recommendation from the literature [24], the value of k is set to 100 to ensure model stability and prediction accuracy.

To address uncertainties related to transportation demand, transportation time, and transfer time, this study developed three independent random forest models. Each model is trained on k bootstrap samples to generate k CARTs, resulting in a total of $3 \times k$ decision trees, thereby establishing a comprehensive surrogate model system.

2) *Training and Updating of Surrogate Model*: Figure 1 depicts the training and updating process of the surrogate model. It approximates uncertainty in multimodal transport networks and thereby reduces dependence on Monte Carlo simulations.

First, representative route data are generated based on the existing multimodal transport network structure as the foundation for analysis. To efficiently utilize the historical data, a dynamically updated data pool is constructed, with an online filling mechanism to ensure data diversity and non-redundancy, thereby avoiding computational burdens caused by repeated simulations. Subsequently, key features such as decision routes, transportation time, and transfer time are extracted from the data pool, forming a set of feature vectors, as shown in Fig. 2. These feature vectors characterize the intrinsic attributes and potential performance of the routes.

In the subsequent simulations, once the data-pool capacity exceeds the training threshold, the feature-vector set is used

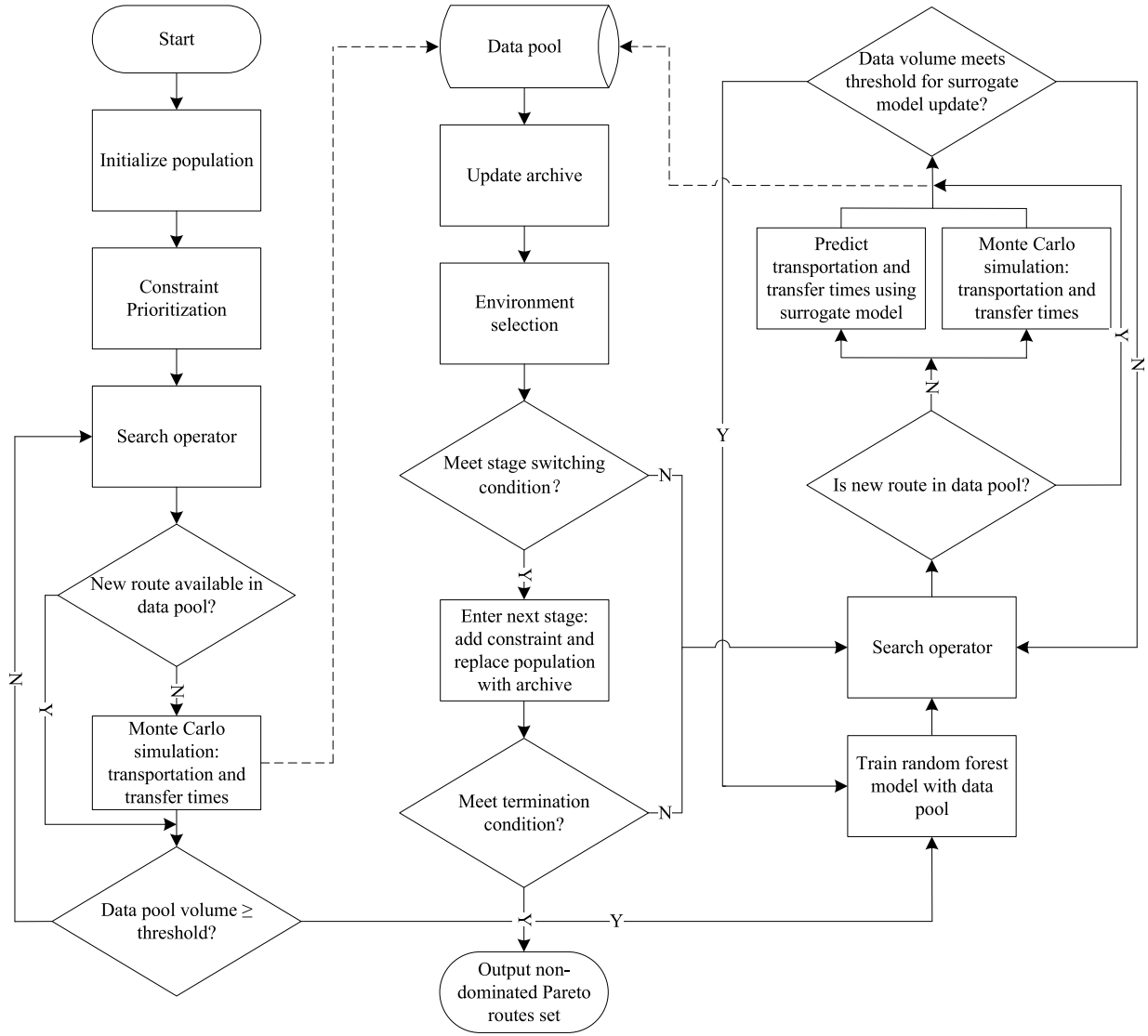


Fig. 1. The data-driven and constraint prioritization-based evolutionary algorithm framework. The algorithm iterates as a closed loop comprising the data pool, surrogate model, and Monte Carlo simulation. Under uncertainty in freight demand and travel time, population evolution is guided by a constraint-priority strategy. When the data volume reaches the surrogate-update threshold, the random-forest surrogate is retrained. Newly generated routes are validated via simulation and, if accepted, appended to the data pool. When the stage switching criteria is met, the archive set is used to reinitialize the population. The cycle repeats until a termination criterion is satisfied, yielding non-dominated Pareto multimodal plans.

as training data to construct a random-forest-based surrogate model. By learning patterns from historical data, the model efficiently predicts uncertainties in new routes (such as transportation demand, transportation time, and transfer time), allowing for rapid performance evaluation. Each new route is first checked for its presence in the data pool: if it exists, its function values are directly mapped using historical data; if it does not, a probabilistic mechanism is introduced: half of the probability uses the surrogate model to speed up the prediction, and half of the probability continues to use Monte Carlo simulation to ensure the accuracy of the results.

Finally, when the data pool reaches the update threshold, the surrogate model is retrained with the newly added samples to improve its generalization and prediction accuracy. If the threshold is not reached, new routes are generated using the reproduction operator to continuously optimize the transportation strategy.

C. Stage Switching Criteria

In the proposed algorithm, the evolutionary process is naturally divided into multiple stages based on the number of constraints, making the timing of stage transitions crucial. To address this, a stage switching criteria is developed, relying on variations in objective function values across populations.

First, the objective function values of the current and previous populations are normalized. The normalization process involves determining the ideal point and the nadir point. The ideal point is $\mathbf{z}^* = \{z_1^{\min}, z_2^{\min}, \dots, z_m^{\min}\}$, and the nadir point is $\mathbf{z}^{\text{nad}} = \{z_1^{\max}, z_2^{\max}, \dots, z_m^{\max}\}$, where $z_i^{\min}$ and $z_i^{\max}$ are the minimum and maximum values of the i -th objective in P . Subsequently, the objective vector of the solution $\mathbf{x}$ is normalized as $\mathbf{F}'(\mathbf{x}) = (f'_1(\mathbf{x}), f'_2(\mathbf{x}), \dots, f'_m(\mathbf{x}))$ using the following formula:

$$f'_i(\mathbf{x}) = \frac{z_i^{\max} - f_i(\mathbf{x})}{z_i^{\max} - z_i^{\min}}, i = 1, 2, \dots, m \quad (29)$$

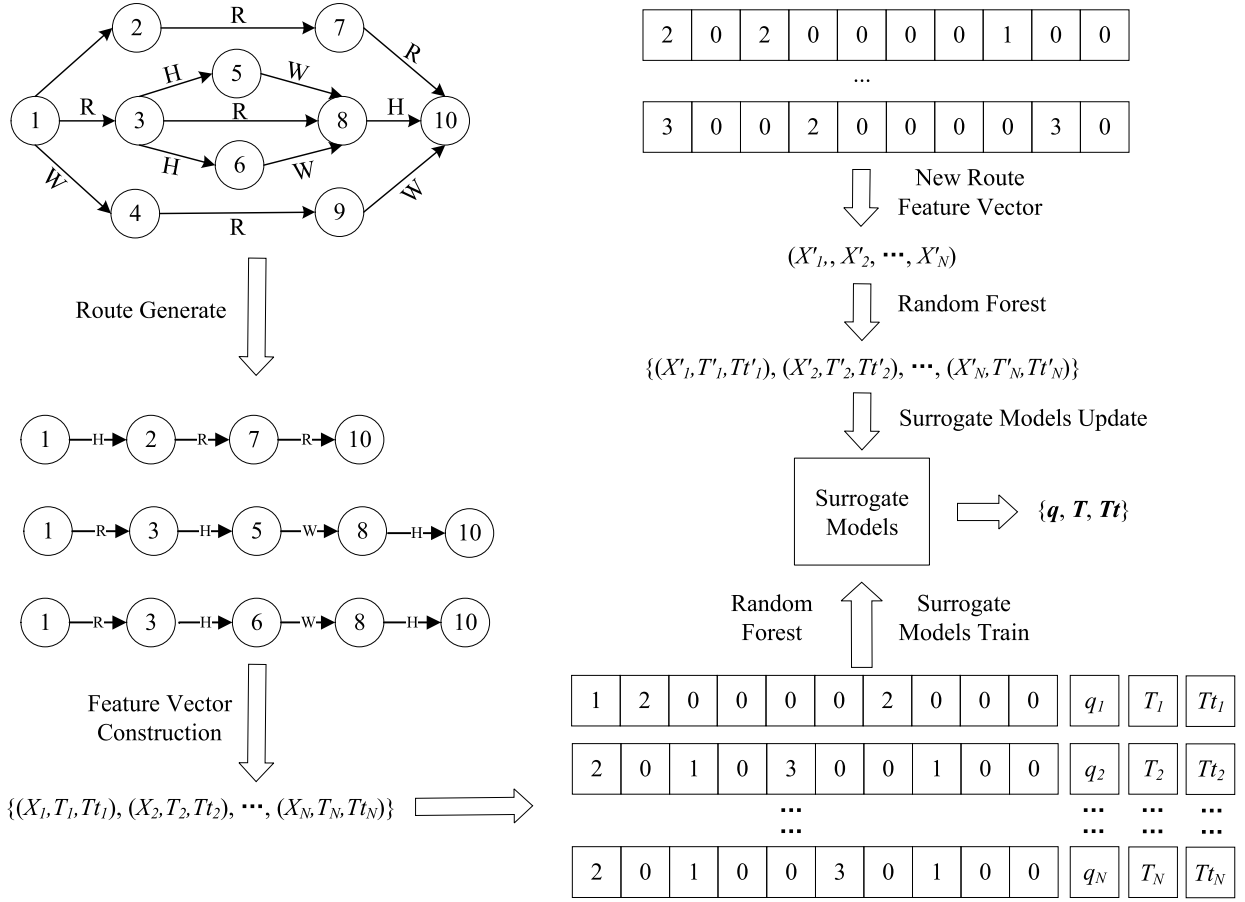


Fig. 2. Construction of feature vectors, training of random forest models, and subsequent updates.

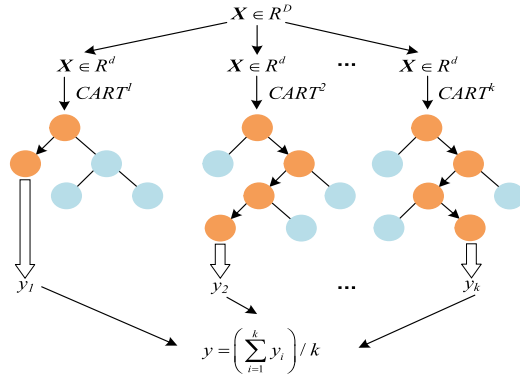


Fig. 3. Illustration of a random forest model: a comprehensive diagram showcasing the structure and mechanism of a random forest, consisting of multiple decision trees trained on bootstrap samples for accurate prediction and modeling of complex data.

Then, the absolute difference between the normalized values is calculated. If this difference falls below a predefined threshold Δ , the evolutionary process switches into the next stage. The adaptive threshold is defined by the following formula:

$$\Delta = \frac{\sum_{j=1}^N \sum_{i=1}^m (f'_i(x_j))}{mN} \quad (30)$$

where m is the number of objectives and N is the population size. Δ represents the average objective value of all solutions

across each objective dimension. As the number of objectives increases, the threshold adjusts accordingly. Since the true feasible PF of a CMOP with m objectives is typically an $(m - 1)$ -dimensional manifold, a fixed threshold may not be suitable for CMOPs with varying objectives, highlighting the need for an adaptive threshold.

D. Constraint Prioritization Strategy

In constrained multi-objective optimization, the true PF of complex multiobjective problems can be shaped by the unconstrained PF and the feasible region defined by all constraints. Building on this insight, we analyze the influence and mechanisms of each constraint on the PF to establish a scientific priority for the constraints. This method aims to guide the optimization algorithm in effectively balancing exploration and exploitation, facilitating efficient solutions to multiobjective optimization problems with complex constraints.

First, the PF is estimated by randomly initializing the population, disregarding constraints (19)–(28). Based on the method described in Section III-C, a check is made to determine if the transition criteria are met. If it is, the priority of each constraint is determined according to the current evolutionary state of the population. Next, the ratio of feasible solutions among the non-dominated solutions in P is calculated. A lower ratio of feasible solutions under a given constraint indicates that the

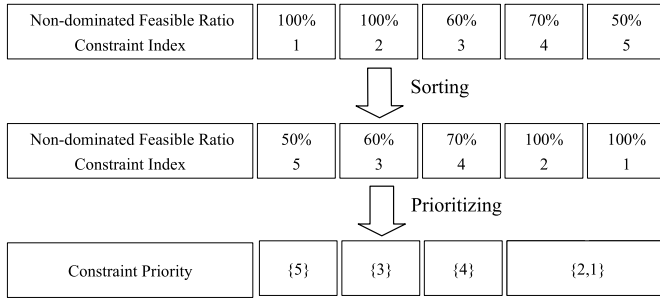


Fig. 4. An example illustrating the prioritization of constraints based on the feasible ratio of non-dominated solutions.

constraint has a greater impact on the unconstrained PF (UPF), thereby assigning it a higher priority. Notably, if all solutions in the population are feasible under current constraints, they are likely to remain feasible under the remaining constraints as well. To enhance search efficiency, the remaining constraints are processed together in a batch.

To illustrate the constraint-handling technique more clearly, an example is provided in Fig. 4. In the initial population, the feasibility ratios of non-dominated solutions under various constraints in the CMOP are unordered. First, we sort the feasibility ratios of each constraint in ascending order and update their corresponding constraint indices. Finally, based on the sorted indices, the constraint priorities are determined as {5}, {3}, {4}, {2, 1}.

Algorithm 2 Constraint Prioritization

Input: P (main population), Δ (stage switching threshold)

Output: $Prio$ (constraint priority)

```

1:  $P \leftarrow$  Randomly initialize  $N$  solutions
2:  $Prio \leftarrow$  Initial_Index()
3: Evaluate  $P$  by MOP
4: while termination condition is not met do
5:    $O \leftarrow$  Reproduction_Operator( $N$ )
6:    $P \leftarrow P \cup O$ 
7:   Evaluate  $P$  by current_cons
8:    $P \leftarrow$  Environmental_Selection( $P, N, current\_cons$ )
9:   if  $Max\_change \leq \Delta$  then
10:    /* stage transition criteria is met */
11:     $P' \leftarrow$  Identify the non-dominated solutions
12:     $fr \leftarrow$  Calculate feasibility ratio of solutions of  $P'$ 
13:     $Prio \leftarrow$  Prioritize each constraint based on the  $fr$ 
14:   end if
15: end while
16: Return  $Prio$ 

```

E. Search Operator

In a constrained multiobjective evolutionary algorithm (CMOEA), the search operator, as a key component, plays a crucial role in generating high-quality offspring solutions. Typically, designing an effective operator ensuring convergence while maintaining diversity. To this end, we propose and combine two proven successful differential evolution (DE) operators [25] to achieve a balance between convergence and diversity. The detailed description is as follows:

1) DE/rand-to-best/1/bi:

$$\mathbf{v}_i = \mathbf{x}_{r_1} + \text{rand}(\mathbf{x}^* - \mathbf{x}_{r_1}) + F(\mathbf{x}_{r_2} - \mathbf{x}_{r_3}) \quad (31)$$

$$\mathbf{u}_{i,j} = \begin{cases} \mathbf{v}_{i,j}, & \text{if } \text{rand}_j < CR \text{ or } j = j_{\text{rand}} \\ \mathbf{x}_{i,j}, & \text{otherwise} \end{cases} \quad (32)$$

1) DE/current-to-rand/1:

$$\mathbf{u}_i = \mathbf{x}_i + \text{rand}(\mathbf{x}_{r_1} - \mathbf{x}_i) + F(\mathbf{x}_{r_2} - \mathbf{x}_{r_3}) \quad (33)$$

Here, $i \in \{1, \dots, N\}$, where N represents the population size, and $j \in \{1, \dots, D\}$, where D denotes the dimension of the decision variables. $\mathbf{x}_i$, $\mathbf{v}_i$, and $\mathbf{u}_i$ represent the target vector, mutant vector, and trial vector, respectively. $\mathbf{x}_{r_1}$, $\mathbf{x}_{r_2}$, and $\mathbf{x}_{r_3}$ are randomly selected from the population P and are distinct from $\mathbf{x}_i$. $\mathbf{x}^*$ represents the best solution within the current population. The parameters F and CR denote the scaling factor and the crossover control parameter, respectively. The random number rand follows a uniform distribution ranging from 0 to 1, while rand_j is a random number generated from a uniform distribution for each dimension $j \in \{1, \dots, D\}$. The random integer j_{rand} takes values between 1 and D . Notably, the DE/current-to-rand/1 operator exhibits rotation invariance, as binary crossover is not implemented in this operator.

On one hand, the DE/rand-to-best/1/bi operator (as shown in Eqs. (31)–(32)) enhances convergence by effectively leveraging the best solution within the current population. On the other hand, the DE/current-to-rand/1 operator (see Eq. (33)) increases population diversity by incorporating three randomly selected solutions. To enhance implementation efficiency, these two offspring generation operators are selected with equal probability, thus achieving a balance between convergence and diversity. Additionally, parameters F and CR are randomly drawn from the scaling factor pool $\{0.6, 0.8, 1.0\}$ and the crossover control parameter pool $\{0.1, 0.2, 1.0\}$, respectively, following relevant literature recommendations [26].

Selecting the best solution from an approximate Pareto set is crucial for balancing objective optimization and constraint satisfaction. The best solution is identified based on the objective function values and constraints of the entire population. In extreme scenarios—where the population consists solely of feasible or infeasible solutions—a non-dominated solution is randomly selected to enhance search diversity. In typical cases—where both feasible and infeasible solutions coexist—a non-dominated feasible solution is chosen to enhance constraint satisfaction. Detailed implementation of the search operator is provided in Algorithm 3.

IV. EXPERIMENTAL STUDIES

To validate the effectiveness of the proposed DD-CPEA and the constructed model for multimodal transportation route optimization under hybrid uncertain environments, we designed three test cases of different scales: small, medium, and large. All test cases involve three transportation modes: road, rail, and waterway.

A. Test Case

As analyzed in Section II-B, the uncertain transportation demand for freight is represented by triangular fuzzy numbers,

Algorithm 3 Search Operator**Input:** P (main population), N (population size)**Output:** O (offspring population)

```

1:  $O \leftarrow \emptyset$ 
2:  $fr \leftarrow$  Calculate the rate of feasible solutions
3: for  $i \in 1:N$  do
4:   Randomly choose a  $F$  value from  $\{0.6, 0.8, 1.0\}$ 
5:   Randomly choose a  $CR$  value from  $\{0.1, 0.2, 1.0\}$ 
6:   if  $rand < 0.5$  then
7:     if  $fr = 0$  or  $fr = 1$  then
8:       Randomly select  $x^*$  from non-dominated solutions
9:     else
10:      Randomly select  $x^*$  from feasible solutions
11:    end if
12:     $x_{r1}, x_{r2}, x_{r3} \leftarrow$  Randomly choose solutions from  $P$ 
13:     $U_i \leftarrow$  Generate the trial vector by Eqs. (31) and (32)
14:  else
15:     $x_{r1}, x_{r2}, x_{r3} \leftarrow$  Randomly choose solutions from  $P$ 
16:     $U_i \leftarrow$  Generate the trial vector by Eq. (33)
17:  end if
18: end for
19:  $O \leftarrow O \cup U_i$ 
20: Return  $O$ 

```

TABLE III

FUZZY TRANSFER TIMES FOR TRANSSHIPMENT

Transportation Mode	Road	Rail	Water
Road	(0,0,0,0)	(1,1.33,1.67,2)	(1,1.33,1.67,2)
Rail	(1,1.33,1.67,2)	(0,0,0,0)	(2,2.67,3.33,4)
Water	(1,1.33,1.67,2)	(2,2.67,3.33,4)	(0,0,0,0)

with $\tilde{q}$ values of (120t, 150t, 180t), where ‘t’ denotes metric tons. The transportation times between arcs and the transfer times at nodes are characterized by trapezoidal fuzzy numbers, with their specific values provided in Table S-I of the supplementary file and Table III, respectively. Furthermore, Table S-I of the supplementary file outlines the transportation distances and maximum capacities for each arc. The transportation speeds for the three modes are 76, 60, and 30 km/h, respectively. Furthermore, the transportation costs for the three modes are 0.3, 0.2, and 0.1 RMB per kilometer, respectively. Liang et al. [27] conducted a statistical analysis of carbon emission factors for road, rail, and water freight from 2011 to 2019. The average carbon emission factors over the nine-year period were calculated to be 0.05771, 0.00820, and 0.00911 kg/t · km, respectively. Table IV presents the transshipment of carbon emission factors [28] and unit costs [29] across different modes of transportation. The hard time window for goods arrival is defined with a lower bound of 0 hours and an upper bound of 100 hours. For soft time windows, any arrival outside this range incurs penalties: 30 RMB/(t · h) for early arrivals and 50 RMB/(t · h) for late arrivals. Non-compliance with the hard time window renders the transportation plan infeasible. The confidence level for the chance constraints in the objective function is set at 80%, with parameters α and β both assigned a value of 0.8.

TABLE IV

CARBON EMISSIONS FACTORS AND UNIT COSTS FOR TRANSSHIPMENT

Transportation Mode	Road	Rail	Water
Road	0/0	0.128/3.09	0.117/5.23
Rail	0.128/3.09	0/0	0.113/26.62
Water	0.117/5.23	0.113/26.62	0/0

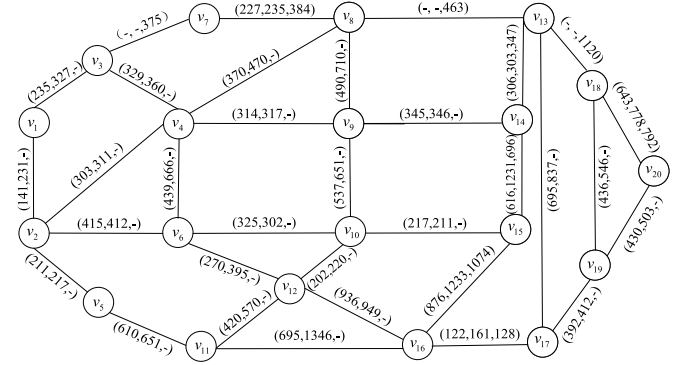


Fig. 5. Multimodal transportation test network with 20 nodes. For example, (141, 231, -) represents the transportation distances between nodes v_1 and v_2 for road, rail, and waterway, respectively, where “-” indicates that the corresponding mode is unavailable. Unit: kilometers.

For a small-scale test case involving 20 nodes, Fig. 5 provides the distance information between each pair of nodes under various transportation modes. Notably, certain transportation modes may be unavailable between specific node pairs, indicated by a “-”. Table S-II of the supplementary file presents the maximum transportation capacity between node pairs. Table S-III of the supplementary file provides the mixing time windows for a multimodal transportation network with 20 nodes. The maximum transportation budget is 130,000 RMB, and the carbon emission threshold is limited to 10,000 kg. Similarly, for the medium-scale case study with 50 nodes, the maximum transportation budget is 440,000 RMB, and the maximum carbon emission threshold is 44,000 kg; for the large-scale case study with 100 nodes, the maximum transportation budget is 1,100,000 RMB, and the maximum carbon emission threshold is 100,000 kg. For the medium-scale and large-scale test cases, information such as transportation distance, transportation capacity, and fuzzy transportation time is provided in Tables S-IV, S-VII, and S-VIII of the supplementary file.

B. Performance Metric

In practical applications of constrained multiobjective optimization, the lack of prior knowledge about the constrained Pareto front (CPF), we adopt a widely used metric, namely hypervolume (HV) [30]. This metric can be used to measure the convergence and diversity of the solution set by setting a nadir point. For small, medium, and large-scale test cases, we set nadir points of (170,000, 11,000, 200), (450,000, 45,000, 400) and (1,200,000, 110,000, 2,200), respectively. A larger HV value indicates better convergence and diversity of the solution set. Notably, solutions that violate hard constraints hold no significance for decision-makers; therefore, only feasible solutions are considered in the evaluation of the HV metric.

C. Parameter Setting

To evaluate the performance and solution efficiency of the proposed DD-CPEA algorithm on the developed multimodal transportation multi-objective optimization model, five state-of-the-art CMOEAs were selected for comparative analysis across small-, medium-, and large-scale test cases.

1) *CMOEAs Under Comparison*: Five promising CMOEAs were involved in the experiments, namely ICMA [31], CCMO [32], MSCMO [33], CMOEA-SDE [34], and DPPPS [35]. Most of these algorithms have demonstrated strong performance in finding non-dominated feasible solutions for CMOPs. A brief introduction to these five CMOEAs is provided below:

- *ICMA (2022)*: This indicator-based evolutionary algorithm for CMOPs uses cost-value-based distance and constraint violation metrics to guide the population in exploring promising regions uniformly.
- *CCMO (2021)*: A co-evolutionary framework was proposed to solve complex CMOPs by optimizing an auxiliary problem and sharing offspring information between two populations.
- *MSCMO (2021)*: This multi-stage evolutionary algorithm introduces constraints progressively and accelerates convergence by prioritizing constraint handling.
- *CMOEA-SDE (2023)*: The algorithm balances objective optimization and constraint satisfaction through a strictly constrained dominance relation and a constrained shift-based density estimation strategy.
- *DPPPS (2022)*: A three-population co-evolutionary algorithm was proposed, balancing objective functions and constraints using weak co-evolutionary relationships and constraint relaxation.

2) *Other Parameters*: The population size N was set to 100. To ensure algorithm convergence, the maximum number of function evaluations was set to 10,000 for various test cases. For reproduction, ICMA employed the DE operator [36] to generate offspring. CCMO, MSCMO, CMOEA-SDE, and DPPPS employed simulated binary crossover (SBX) [37] and polynomial mutation [38], with the crossover probability set to 1 and the mutation probability set to $1/D$ (where D represents the number of decision variables in the CMOP). The distribution indices for both operators were set to 20. For statistical analysis, all algorithms were independently run 30 times and Friedman and Nemenyi statistical tests [39] were performed. Additionally, to ensure a fair comparison, the optimal parameters for comparison of CMOEAs under evaluation were kept consistent with those in the original study.

D. Experimental Results

In this section, we utilize the chance-constrained programming model developed in Section II-E to conduct comparative experiments between the proposed DD-CPEA algorithm and five state-of-the-art CMOEAs. The experiments are carried out on both small-scale and large-scale test cases, aiming to validate the effectiveness and performance of the proposed algorithm.

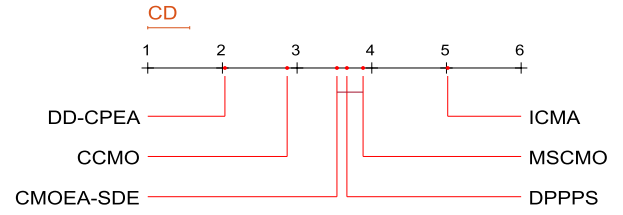


Fig. 6. Friedman rankings of six algorithms based on HV metrics for the 20-node test case. The results reveal significant differences among the algorithms, except for CCMO, DPPPS, MSCMO, and ICMA, where no significant differences were observed.

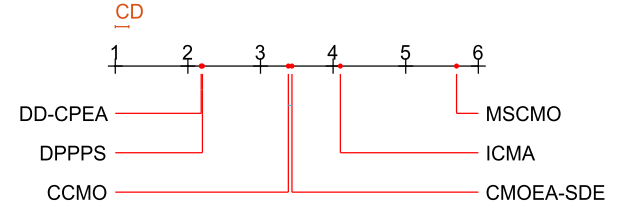


Fig. 7. Friedman rankings of six algorithms based on HV metrics for the 50-node test case. The results reveal significant differences among the algorithms, except for DD-CPEA and DPPPS as well as CCMO and CMOEA-SDE, where no significant differences were observed.

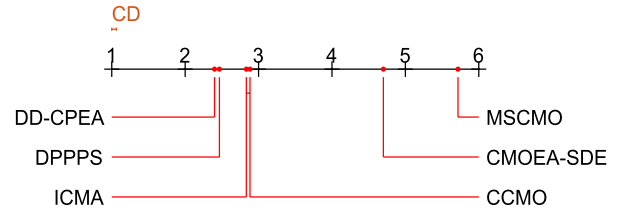


Fig. 8. Friedman rankings of six algorithms based on HV metrics for the 100-node test case. The results reveal significant differences among the algorithms, except for ICMA and CCMO, where no significant differences were observed.

Table V summarizes the HV metric results for all algorithms across small-scale, medium-scale, and large-scale test cases. A detailed analysis reveals that the proposed DD-CPEA consistently achieves the highest mean HV value among all comparison algorithms in all test cases, underscoring its superior performance. This is mainly attributable to the constraint prioritization strategy employed by DD-CPEA. By introducing complex constraints one by one into the optimization process, this strategy effectively reduces the difficulty of solving the original problem. Notably, in the 50-node test case, MSCMO failed to obtain a feasible solution set. This may be because the algorithm became trapped in a local optimum during the search and did not sufficiently converge toward the vicinity of the Pareto front.

Furthermore, an examination of the stability statistic, standard deviation (STD), shows that DD-CPEA has the lowest standard deviation. This finding not only reinforces DD-CPEA's performance superiority but also indicates greater stability and reliability when addressing multimodal transport networks of varying scales compared to other peer algorithms.

Fig. 6, 7, and 8 present the Friedman rankings based on the HV metric for the proposed DD-CPEA algorithm and five comparison CMOEAs on small-, medium- and large-scale test cases. The results of the Friedman and Nemenyi

TABLE V
COMPARISON OF HV METRIC VALUES (MEAN AND STANDARD DEVIATION) FOR THE PROPOSED DD-CPEA AND FIVE
PEER ALGORITHMS. THE BEST-PERFORMING METRIC ACROSS ALL ALGORITHMS IS HIGHLIGHTED

Network Scale	Metric	ICMA	CCMO	MSCMO	CMOEA-SDE	DPPPS	DD-CPEA
20 nodes	MEAN	2.8324e-1	2.9247e-1	2.8644e-1	2.8675e-1	2.8865e-1	2.9412e-1
	STD	1.45e-2	2.80e-3	1.55e-2	1.83e-2	8.27e-3	2.66e-3
50 nodes	MEAN	5.4796e-2	5.6737e-2	–	5.5997e-2	5.9550e-2	6.2134e-2
	STD	1.03e-2	5.36e-3	–	4.99e-3	6.94e-3	2.59e-3
100 nodes	MEAN	1.0132e-2	1.0733e-2	9.3116e-3	8.4657e-3	1.0664e-2	1.2117e-2
	STD	1.96e-3	1.72e-3	1.63e-3	1.59e-3	1.27e-3	8.64e-4

TABLE VI
MAXIMUM AND MINIMUM VALUES OF THE OBJECTIVE FUNCTIONS OBTAINED BY THE PROPOSED DD-CPEA AND FIVE PEER ALGORITHMS

Network Scale	Result	ICMA			CCMO			MSCMO			CMOEA-SDE			DPPPS			DD-CPEA		
		f_1 (km)	f_2 (t)	f_3 (h)	f_1 (km)	f_2 (t)	f_3 (h)	f_1 (km)	f_2 (t)	f_3 (h)	f_1 (km)	f_2 (t)	f_3 (h)	f_1 (km)	f_2 (t)	f_3 (h)	f_1 (km)	f_2 (t)	f_3 (h)
20 node	Min	64097.2	4024.42	77.43	64097.2	4223.19	74.11	64097.2	3734.12	51.56	64097.2	4223.19	78.22	64097.2	4223.19	73.25	64097.2	4023.19	58.88
	Max	97564.03	7964.17	121.34	88898.1	4903.91	121.34	177920.3	5171.37	121.34	93251.25	7787.35	121.34	88898.1	4903.91	121.34	114552.8	5308.59	121.34
50 node	Min	272893.6	33548.43	242.63	288922.1	33930.24	242.63	292767.4	32876.24	106.86	299673.3	33346.27	243.22	272893.6	33346.27	242.63	272893.6	33346.27	242.63
	Max	347016.1	36604.26	284.56	320968.2	36604.26	285.67	594888.1	43816.12	238.76	344352.9	36402.1	255.27	344352.9	36604.26	284.56	344352.9	35935.08	303.48
100 node	Min	1026785	92874.79	1923.21	1020706	91991.7	1929.97	1059637	75925.2	1285.36	1039745	95467.88	1934.65	1012747	93770.96	1914.66	994599.3	87635.67	1941.95
	Max	1092086	97948.82	2171.96	1091034	97146.8	1971.44	35622534	99395.1	2021.06	1072132	99093.12	2027.3	1064991	98920.85	2116.45	1097149	93667.7	2040.84

statistical tests show significant differences among the algorithms, with a p-value of 6.1458e-54. When the p-value is below the significance level (typically 0.05), the observed difference is considered statistically significant. For the small-scale test case, significant differences were found between the six algorithms except for CCMO, DPPPS, MSCMO, and ICMA. For the medium-scale test case, significant differences were observed among all algorithms except between DD-CPEA and DPPPS, and between CCMO and CMOEA-SDE, with a p-value of zero. Similarly, for the large-scale test case, six algorithms showed significant differences except between ICMA and CCMO, with a p-value of zero. Across multimodal transportation networks with 20, 50, and 100 nodes, the proposed DD-CPEA significantly outperformed the other comparative algorithms and achieved the top Friedman ranking.

Table VI provides a detailed summary of the optimal results obtained by all algorithms in small-scale, medium-scale, and large-scale test cases, effectively demonstrating their capabilities in acquiring boundary solutions. Specifically, the table presents performance data for three objective functions, “ f_1 ,” “ f_2 ,” and “ f_3 ,” for each algorithm, with the “Max” and “Min” rows indicating the maximum and minimum values of the obtained solution sets for each objective function. A closer look at Table VI reveals clear differences between the algorithms in reaching the maximum and minimum function values in the final obtained solution sets. Further analysis shows that the proposed DD-CPEA consistently achieved the best maximum and minimum values across all test case scales. DD-CPEA demonstrates a stronger capability in identifying boundary non-dominated solutions compared to other comparison algorithms. This is because, during the sequential introduction of constraints, information from promising infeasible solutions is retained. Such information not only enriches population diversity but also provides crucial search guidance for ultimately approaching the true CPF.

To visually demonstrate the characteristics of the non-dominated feasible solution sets obtained by each algorithm, Fig. 9 shows the obtained solution sets with median HV values over 30 independent runs on the network with 20 nodes. As illustrated in Fig. 9, the proposed DD-CPEA generated a number of non-dominated feasible solutions, which exhibited both good diversity and good convergence. In contrast, the ICMA, CCMO, and CMOEA-SDE produced fewer non-dominated feasible solutions compared to DD-CPEA. While the MSCMO and DPPPS obtained a number of non-dominated solutions, the diversity of the solution sets is relatively poor. For MSCMO, a closer examination of the objective function values reveals suboptimal convergence (i.e., the second objective function value is larger than that of peer algorithms). Similarly, Fig. S-I and S-II of the supplementary file present the obtained solution sets with median HV values over 30 independent runs on the network with 50 nodes and 100 nodes, respectively. Notably, even with the increased complexity of the network, the DD-CPEA algorithm continued to exhibit the best solution distribution and convergence, further validating its superior performance.

Table VII presents the twelve non-dominated feasible solutions generated by the proposed DD-CPEA algorithm in a multimodal transportation network consisting of 20 nodes, along with their corresponding twelve multimodal transportation routing plans. Each routing plan includes three objective function values: total transportation cost, total transportation time, and total carbon emissions, as well as the satisfaction rate with the chance constraints. As shown in the table, the twelve transportation schemes exhibit diverse characteristics in terms of transportation cost, transportation time, carbon emissions, and chance constraint satisfaction rates, offering a variety of alternative solutions for multimodal transportation operators. Decision-makers can select the most suitable scheme based on real-world requirements and their

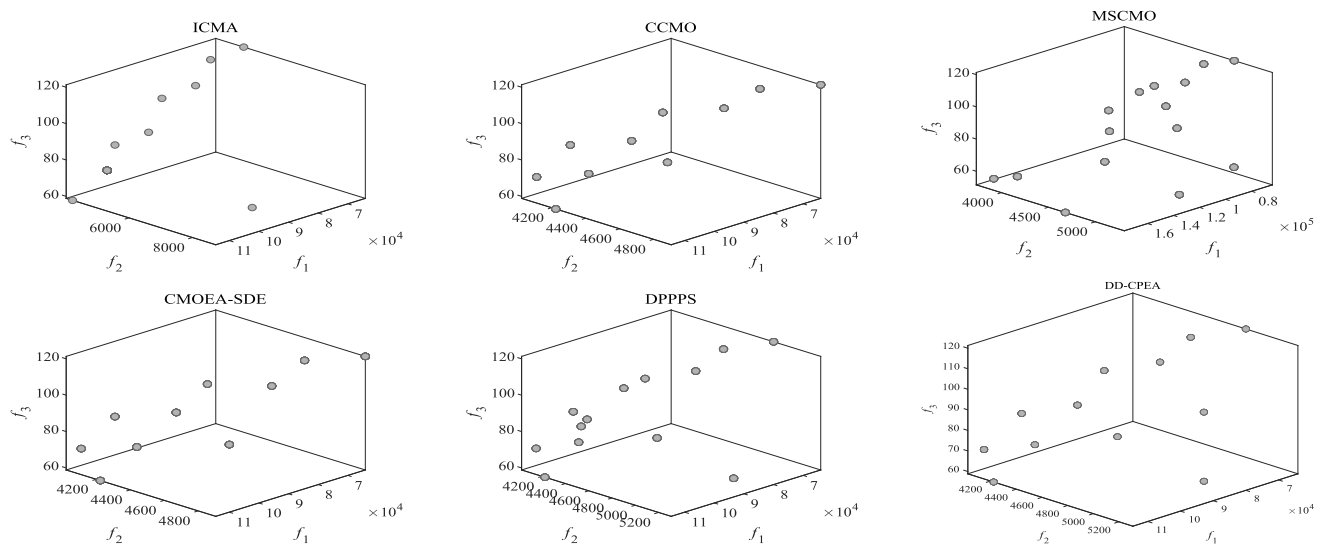


Fig. 9. Nondominated solution set of the proposed DD-CPEA and five comparison algorithms with the median HV result among 30 independent runs on the test case with 20 nodes.

TABLE VII

MULTIMODAL TRANSPORTATION ROUTING PLAN OBTAINED BY THE PROPOSED DD-CPEA ALGORITHM. IN THIS TABLE, ‘H’ DENOTES HIGHWAY IN THE ROAD TRANSPORTATION MODE, ‘R’ REPRESENTS THE RAIL TRANSPORTATION MODE, ‘W’ INDICATES THE WATER TRANSPORTATION MODE, ‘→’ DENOTES THE TRANSPORTATION MODE BETWEEN NODE PAIRS

No.	Multimodal Transportation Routing Plan	Transportation Costs	Transportation Time	Carbon Emissions	Constraint Satisfaction Rate
1	1 $\xrightarrow{R}$ 2 $\xrightarrow{R}$ 4 $\xrightarrow{R}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{R}$ 17 $\xrightarrow{R}$ 19 $\xrightarrow{R}$ 20	109887.20	4030.29	68.48	95%
2	1 $\xrightarrow{R}$ 2 $\xrightarrow{R}$ 6 $\xrightarrow{R}$ 10 $\xrightarrow{R}$ 15 $\xrightarrow{R}$ 16 $\xrightarrow{R}$ 17 $\xrightarrow{R}$ 19 $\xrightarrow{R}$ 20	114552.84	4224.20	58.88	85%
3	1 $\xrightarrow{R}$ 3 $\xrightarrow{R}$ 4 $\xrightarrow{R}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{W}$ 18 $\xrightarrow{R}$ 20	90145.48	4536.58	92.93	100%
4	1 $\xrightarrow{R}$ 3 $\xrightarrow{W}$ 7 $\xrightarrow{W}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{W}$ 18 $\xrightarrow{W}$ 20	64097.20	4903.91	121.34	100%
5	1 $\xrightarrow{R}$ 3 $\xrightarrow{W}$ 7 $\xrightarrow{R}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{W}$ 18 $\xrightarrow{R}$ 20	85779.99	4352.71	101.11	100%
6	1 $\xrightarrow{R}$ 3 $\xrightarrow{W}$ 7 $\xrightarrow{R}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{W}$ 18 $\xrightarrow{W}$ 20	73132.00	4703.92	117.90	100%
7	1 $\xrightarrow{R}$ 3 $\xrightarrow{R}$ 4 $\xrightarrow{R}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{W}$ 18 $\xrightarrow{W}$ 20	77492.00	4887.79	109.62	100%
8	1 $\xrightarrow{R}$ 3 $\xrightarrow{W}$ 7 $\xrightarrow{W}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{R}$ 17 $\xrightarrow{R}$ 19 $\xrightarrow{R}$ 20	88898.10	4223.19	83.32	100%
9	1 $\xrightarrow{R}$ 3 $\xrightarrow{W}$ 7 $\xrightarrow{W}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{W}$ 18 $\xrightarrow{R}$ 20	76580.37	4552.71	104.55	100%
10	1 $\xrightarrow{R}$ 3 $\xrightarrow{W}$ 7 $\xrightarrow{R}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{R}$ 17 $\xrightarrow{R}$ 19 $\xrightarrow{R}$ 20	98097.73	4023.19	79.93	100%
11	1 $\xrightarrow{H}$ 2 $\xrightarrow{R}$ 4 $\xrightarrow{R}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{W}$ 18 $\xrightarrow{R}$ 20	109487.02	5306.98	89.17	93%
12	1 $\xrightarrow{R}$ 3 $\xrightarrow{R}$ 4 $\xrightarrow{R}$ 8 $\xrightarrow{W}$ 13 $\xrightarrow{R}$ 17 $\xrightarrow{R}$ 19 $\xrightarrow{R}$ 20	102463.59	4207.06	71.65	100%

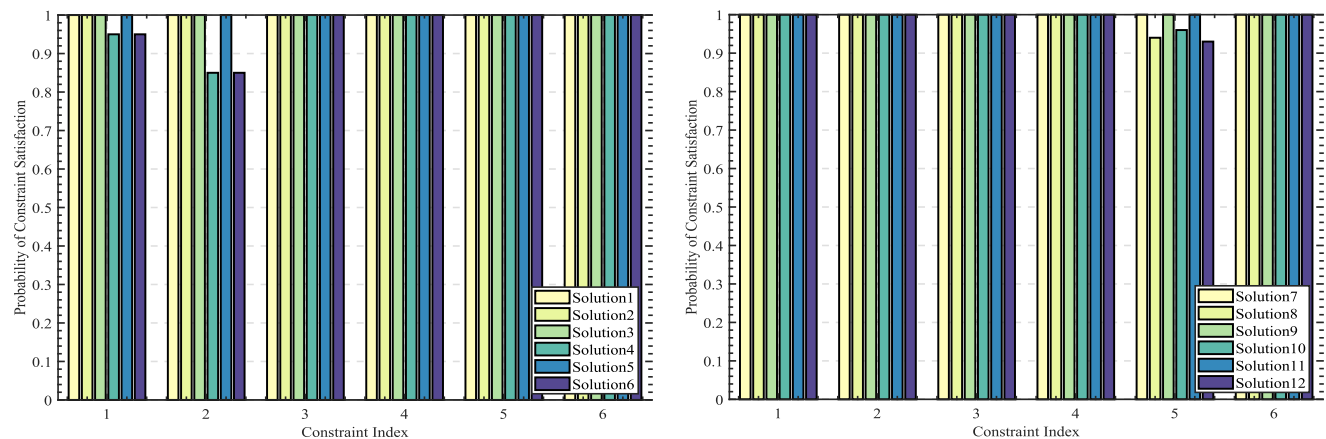


Fig. 10. Constraint satisfaction probabilities of the twelve multimodal transportation routing solutions for different constraint indices.

decision-making preferences to achieve optimal operational outcomes.

Regarding the satisfaction rate of the chance constraints, Table VII shows that all transportation schemes achieve

TABLE VIII

COMPARISON OF THE RUNNING TIME BETWEEN THE BASIC CPEA (I.E., B-CPEA) AND THE DATA-DRIVEN CPEA (I.E., DD-CPEA)

Algorithm \ Network Scale	20	50	100
B-CPEA	1.9312e+2	6.55509e+2	6.3332e+2
DD-CPEA	1.4205e+1	1.3034e+2	2.4021e+2
Time savings	92.64%	80.12%	62.07%

a satisfaction rate above the 80% threshold. More intuitively, Fig. 10 illustrates the satisfaction probabilities of the twelve transportation schemes for each chance constraint within the 20-node multimodal transportation network. Specifically, constraints 1 to 5 correspond to Eq. (22)–(26), while Constraint 6 corresponds to Eq. (28). Serving as a comprehensive constraint, Eq. (28) ensures that the previously mentioned conditions are satisfied across all possible scenarios. As shown in Fig. 9, the transportation schemes derived from the proposed DD-CPEA algorithm significantly exceed the predetermined 80% confidence level regarding satisfaction probabilities for all constraints. This result strongly underscores the effectiveness of the constructed model in solving the chance-constrained programming problem.

In summary, this series of systematic experimental validations convincingly demonstrates that the developed model and its data-driven algorithm framework can efficiently identify multiple Pareto optimal solutions in complex environments characterized by hybrid uncertainties. These optimal solutions comprehensively account for various key factors, including freight transportation demand, transportation time, and transfer time, thereby providing robust support for achieving more efficient and reliable multimodal transportation schemes.

E. Ablation Experiments

As analyzed in the previous section, the Monte Carlo simulation method demands considerable computational resources due to the large number of required samples. To assess the effectiveness of the proposed data-driven approach, we investigated its impact on running time. Specifically, we compared the original CPEA without the data-driven method (referred to as B-CPEA) with the final version of CPEA that incorporates the data-driven method (termed DD-CPEA). As illustrated in Table VIII, the introduction of the data-driven approach led to a substantial reduction in computation time. For networks with 20, 50, and 100 nodes, the time savings were 92.64%, 80.12%, and 62.07%, respectively, resulting in over a 60% reduction across all network scales. This is due to the data-driven approach adopted by DD-CPEA, which effectively reduces the algorithm's reliance on Monte Carlo simulations when handling uncertainty. To provide a more intuitive representation of the comparison of running times, Fig. 11 presents a corresponding bar chart. These results highlight the critical role of the data-driven method in enhancing DD-CPEA efficiency, providing support for further exploration and application of data-driven techniques in addressing similar challenges.

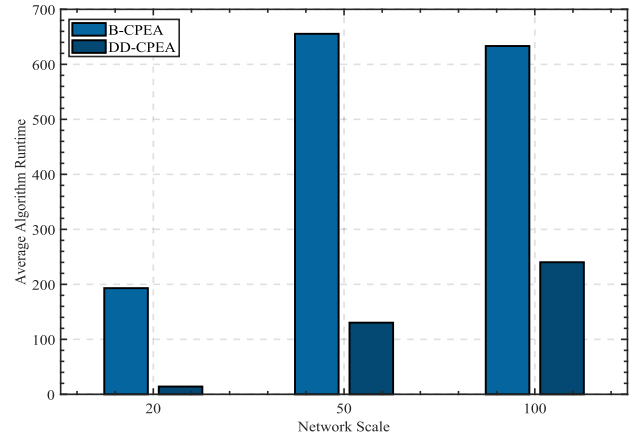


Fig. 11. Comparison of average running times for the Basic CPEA (i.e., B-CPEA) and the Data-Driven CPEA (i.e., DD-CPEA).

TABLE IX

COMPARISON OF THE HV METRIC BETWEEN THE BASIC CPEA (I.E., B-CPEA) AND THE DATA-DRIVEN CPEA (I.E., DD-CPEA)

Algorithm \ Network Scale	20	50	100
B-CPEA	3.0609e-1	6.5362e-2	1.149e-2
DD-CPEA	2.9412e-1	6.2134e-2	1.2117e-2
Percentage difference	3.91%	4.94%	5.17%

To evaluate the impact of the data-driven method on algorithm accuracy, we conducted a detailed statistical analysis of the HV metric variations. As shown in Table IX, the HV metrics of the final Pareto non-dominated solution sets for various scales of multimodal transport networks exhibit minimal differences (within 6%) before and after implementing the data-driven method. This finding underscores that the proposed method significantly enhances algorithm efficiency while maintaining accuracy. Overall, the data-driven method not only optimizes algorithm performance but also demonstrates exceptional stability and reliability in preserving accuracy, providing a strong foundation for the future application of data-driven techniques in complex network optimization problems.

V. CONCLUSION

In this paper, a multiobjective route optimization model based on fuzzy numbers is constructed to address the comprehensive impact of uncertain factors such as freight transportation demand, transportation time, and transfer time under the complex operation background of multimodal transportation. The model incorporates transportation costs, transit time, and environmental factors (carbon emissions) to achieve a comprehensive optimization of economic efficiency, time efficiency, and environmental sustainability. By integrating chance-constrained programming theory, it effectively addresses uncertain variables, thereby enhancing the model's practicality and solvability.

To solve the model, a novel data-driven multiobjective evolutionary optimization method is proposed. By mitigating

the high computational cost associated with traditional Monte Carlo sampling in the presence of multiple uncertain variables, a surrogate model is developed using historical data to predict the values of these uncertain variables, significantly reducing the computational expense of fitness evaluations. Additionally, a constraint prioritization strategy is designed to utilize potential infeasible solutions while prioritizing key constraints, thereby reducing the complexity of the optimization problem and improving solution efficiency.

Through simulation experiments, this study demonstrates the superior performance of the proposed model and algorithm across multimodal transportation networks of varying scales. In small, medium, and large-scale networks, the proposed algorithm consistently outperforms peer algorithms, achieving larger HV values and further underscoring its advantages in complex multimodal transportation route optimization.

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